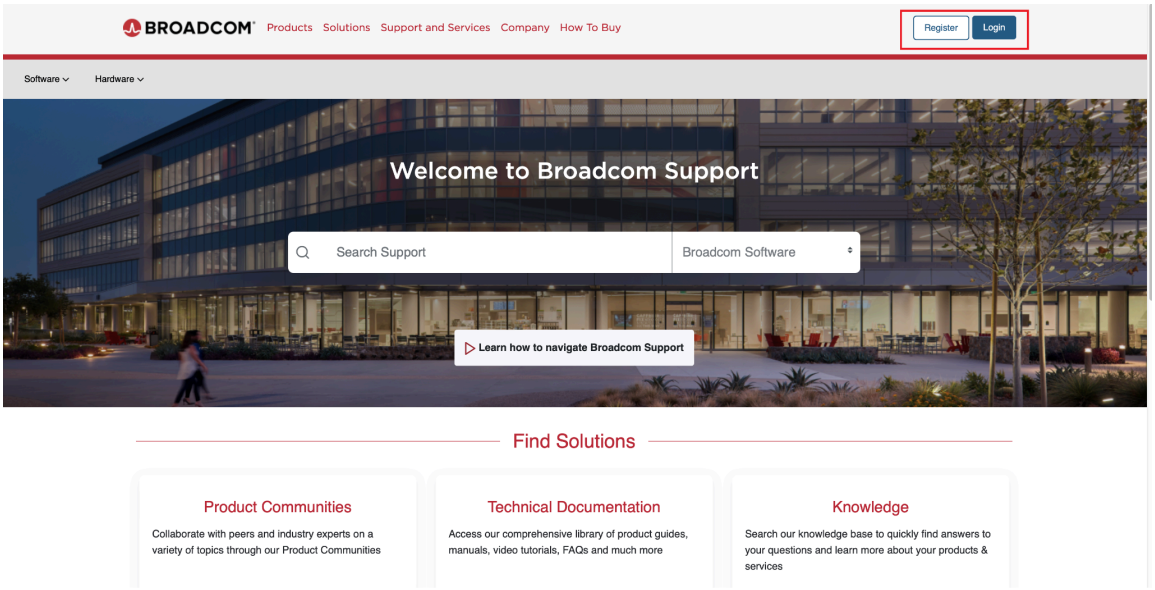
Setting Up VM with VMWare Fusion
Building the VM on Fusion

Step 1: Install VMware Fusion Player

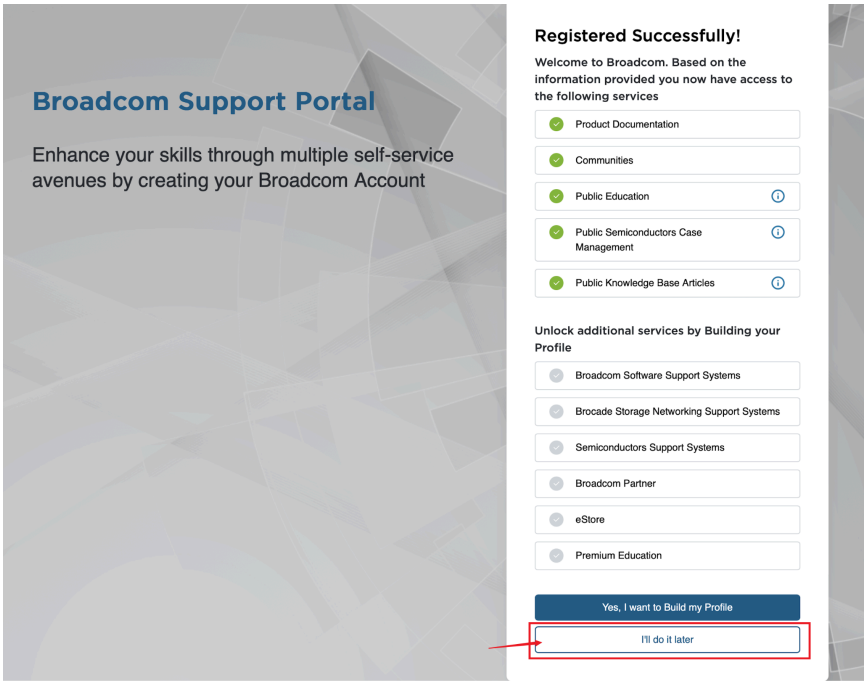
VMware Fusion Player is a free virtual machine software. It is similar to VirtualBox that we have been using, but VirtualBox still cannot run reliably on Apple Silicon machines.

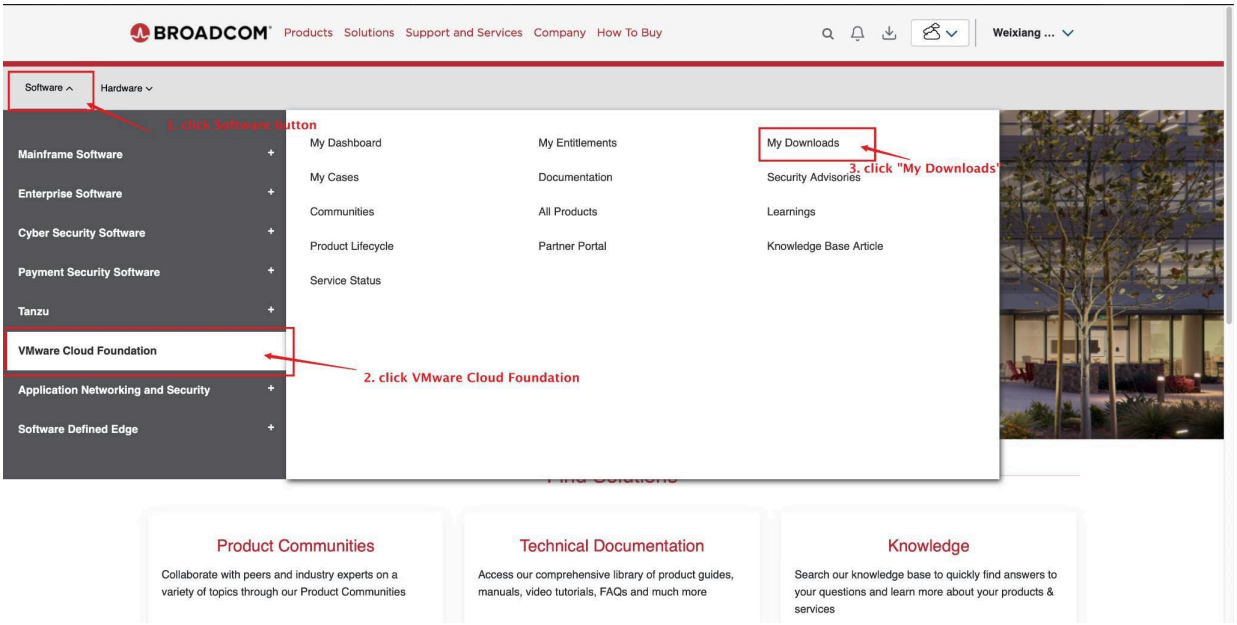
Go to VMware Fusion and either log in or register for a basic Broadcom account.

NOTE: For Windows it is VMware Workstation Pro

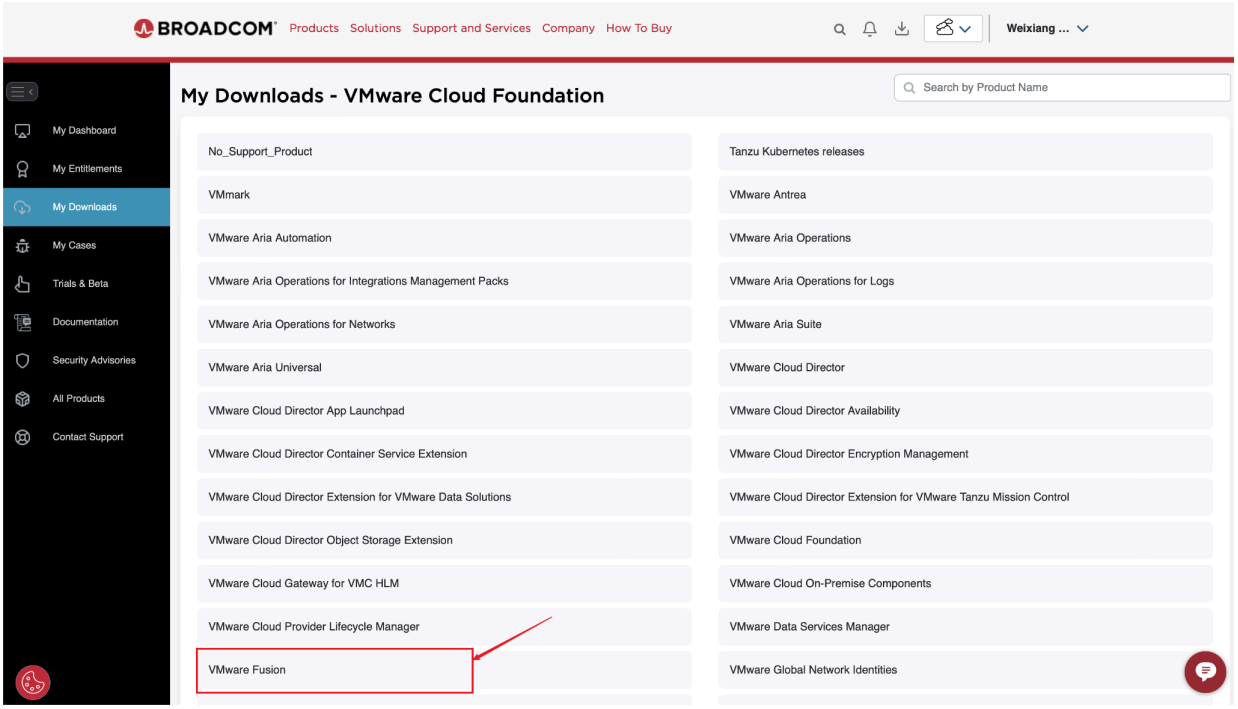


Once logged in the Broadcom account, we come back to the Broadcom Support Portal if we're not redirected there. We click the `Software` dropdown to choose the `VMware Cloud Foundation division` to select `My Downloads`.

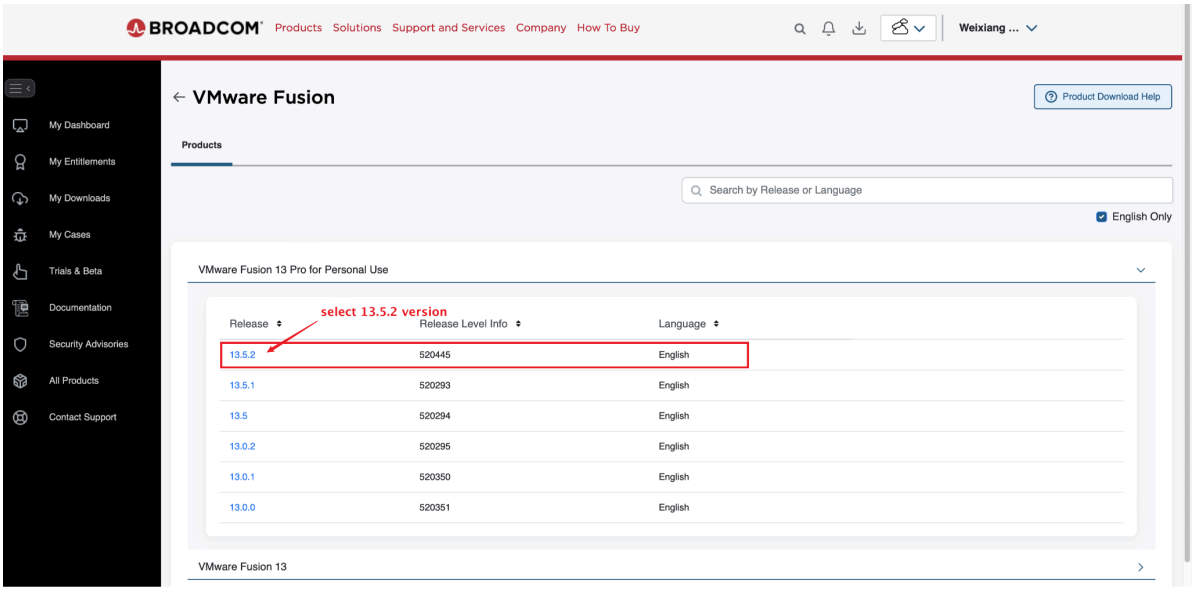




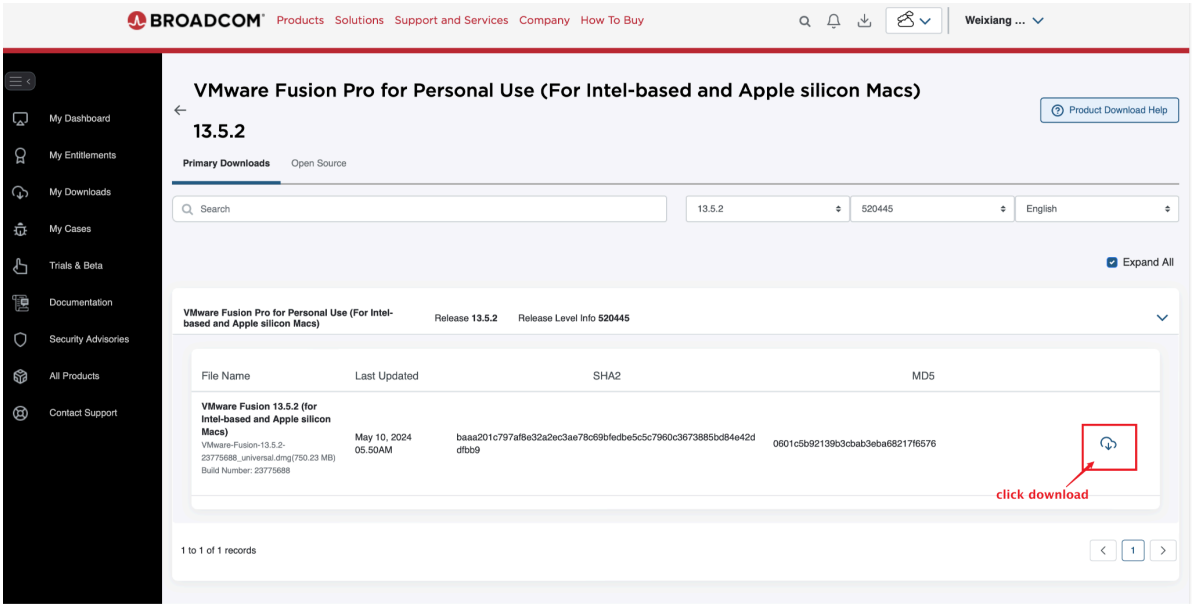
Find `VMware Fusion` in `My Downloads` page.



click `VMware Fusion 13 Pro for Personal Use` dropdown to choose the 13.6.2 version(or later).



In the Primary Downloads page, find the `.dmg` file and click download button to download the dmg file manually.



Installation is straight forward. Double click on the downloaded dmg file and follow the instructions.



After the installation is finished, you can start the VMware Fusion Player.

You will be asked to enter a license key. Select `VMware Fusion Pro for Personal Use`. You will be asked to allow the kernel extensions. Click on `Open Security Preferences`. In the Security & Privacy settings, click on `Allow` to allow the kernel extensions.

NOTE: All the steps remain the same for Windows OS. Just install the new version of the Workstation and rest go on with the setup wizard.

Step 2: Build an Ubuntu VM on VMWare Fusion.

- Step 2.1: Download Ubuntu 22.04 server
- Step 2.2: Create VM on VMWare Fusion
- Step 2.3: Install Ubuntu server
- Step 2.4: Install Ubuntu Desktop

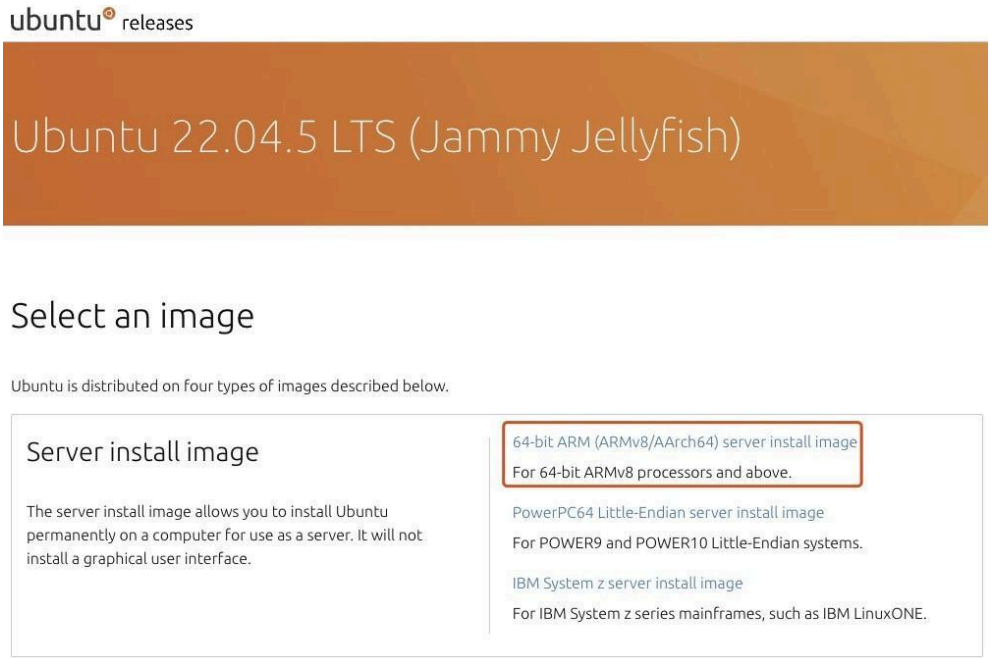
Step 2.1: Download Ubuntu 22.04 server

Since the desktop versions of the Ubuntu 22 version VM for ARM are removed; we will install the Server version instead. We recommend downloading the Ubuntu 22.04 version.

We will download the Ubuntu ISO image first. Go to Ubuntu 22.04.5 and download the Ubuntu 22.04.5 LTS (Jammy Jellyfish) . Make sure you download the `64-bit ARM (ARMv8/AArch64) server install image`. Click on this link for directly downloading iso file

Ubuntu Live Server for Mac

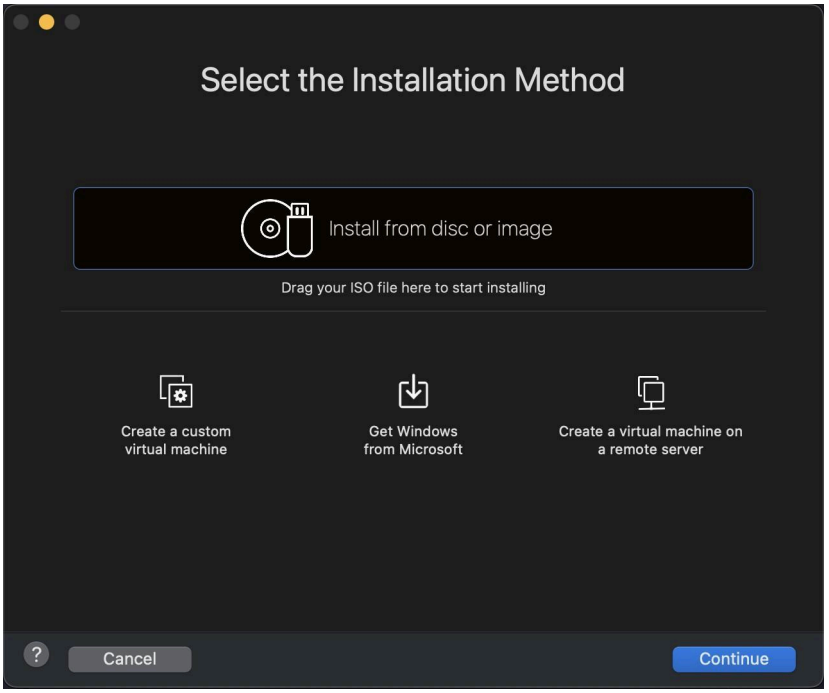
NOTE: For Windows its AMD64 architecture. Click this link to download the iso image
Ubuntu Live Server for Windows



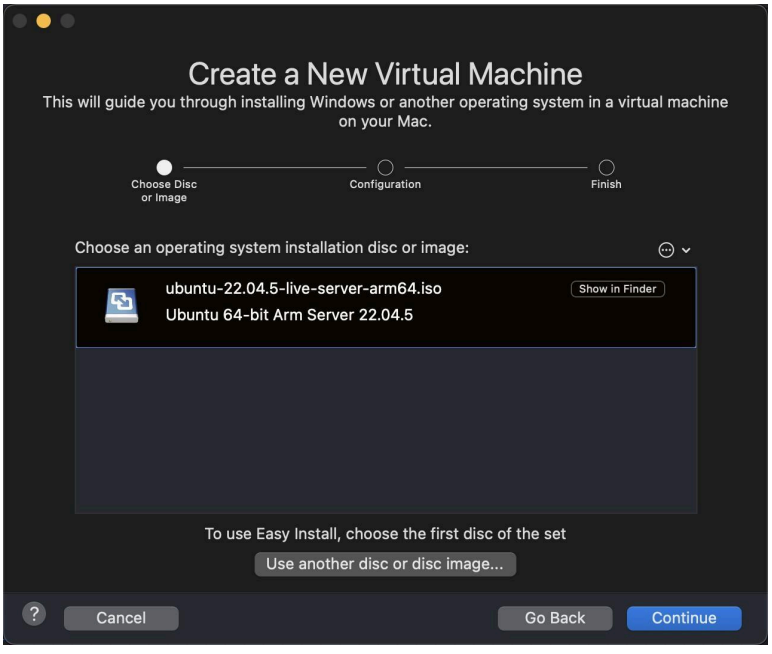
Step 2.2: Create VM on VMWare Fusion/Workstation

After the download is finished, start the VMware Fusion Player. Click on `Create a New Virtual Machine`.

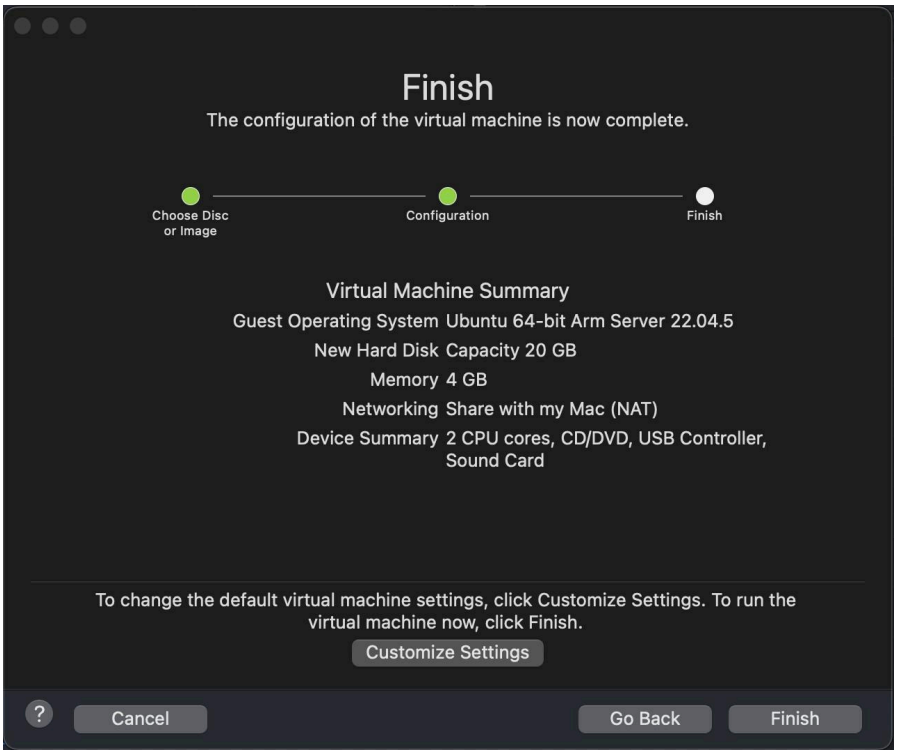
In `Select the Installation Method`, select `Install from disc or image` and click on `Continue`.



Select the downloaded Ubuntu ISO image, and then click on `Continue`.

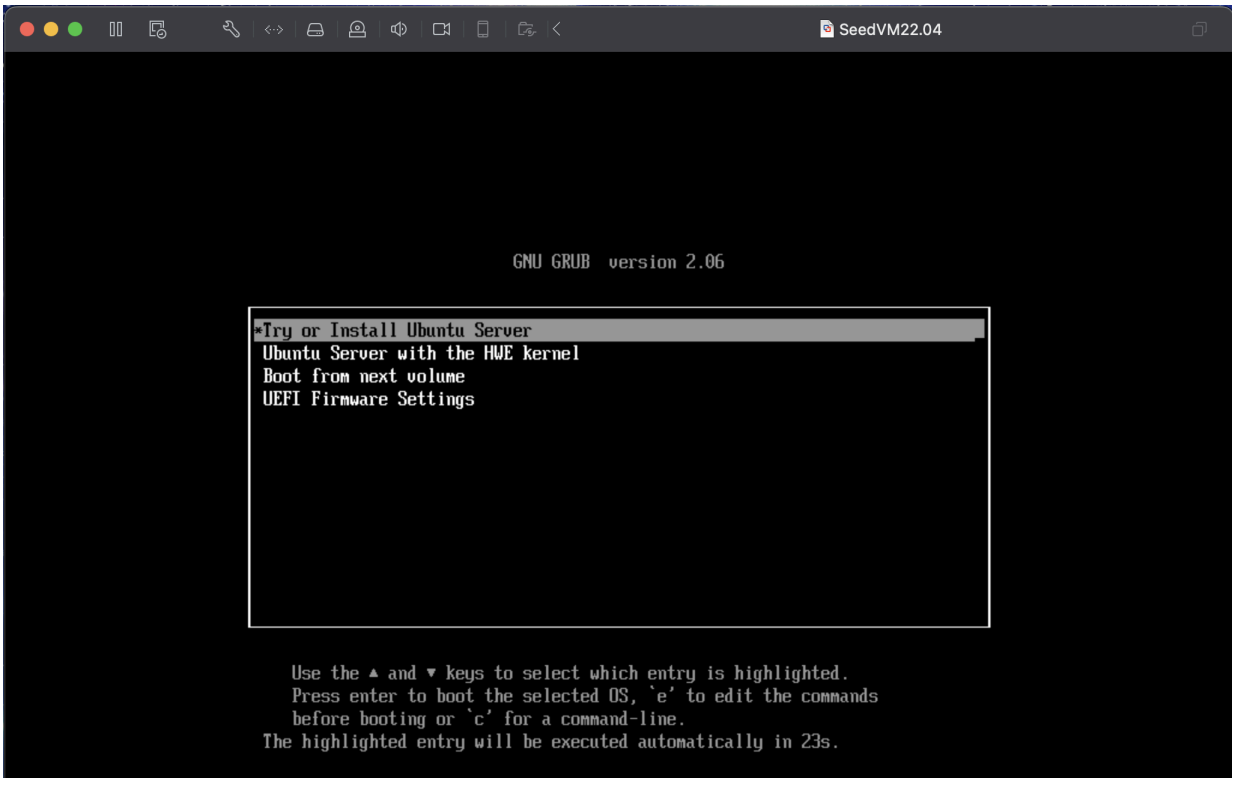


In the next screen, make sure that 2 CPUs and 4 GB of RAM are selected. Click on `Finish`. The VM will be created and started.



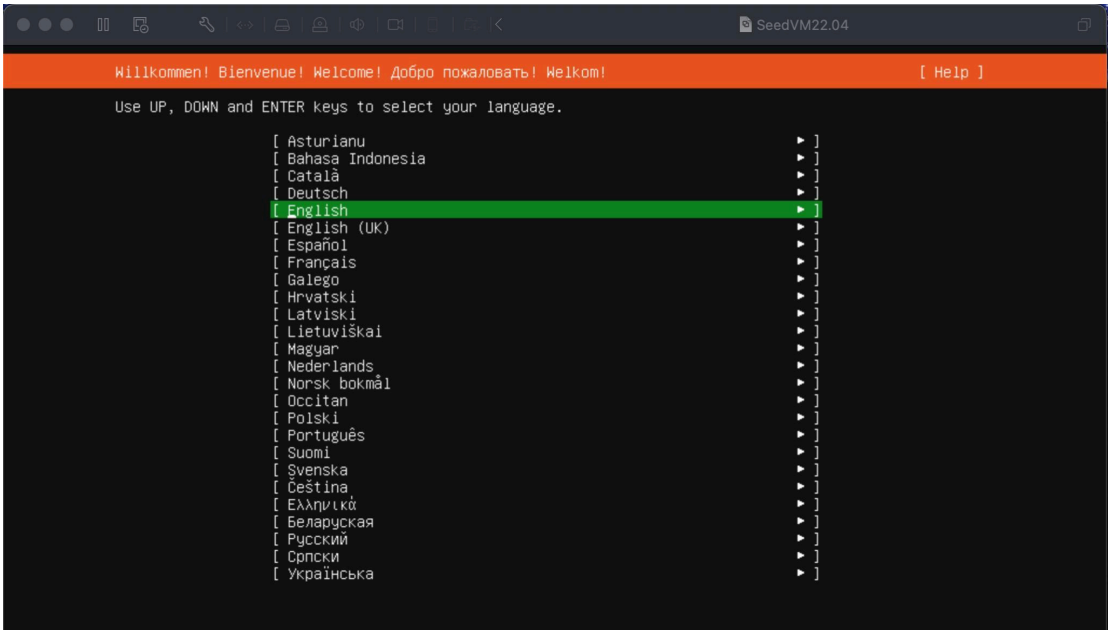
Step 2.3: Install Ubuntu server

After the VM is started, click on `Try or install Ubuntu Server`.

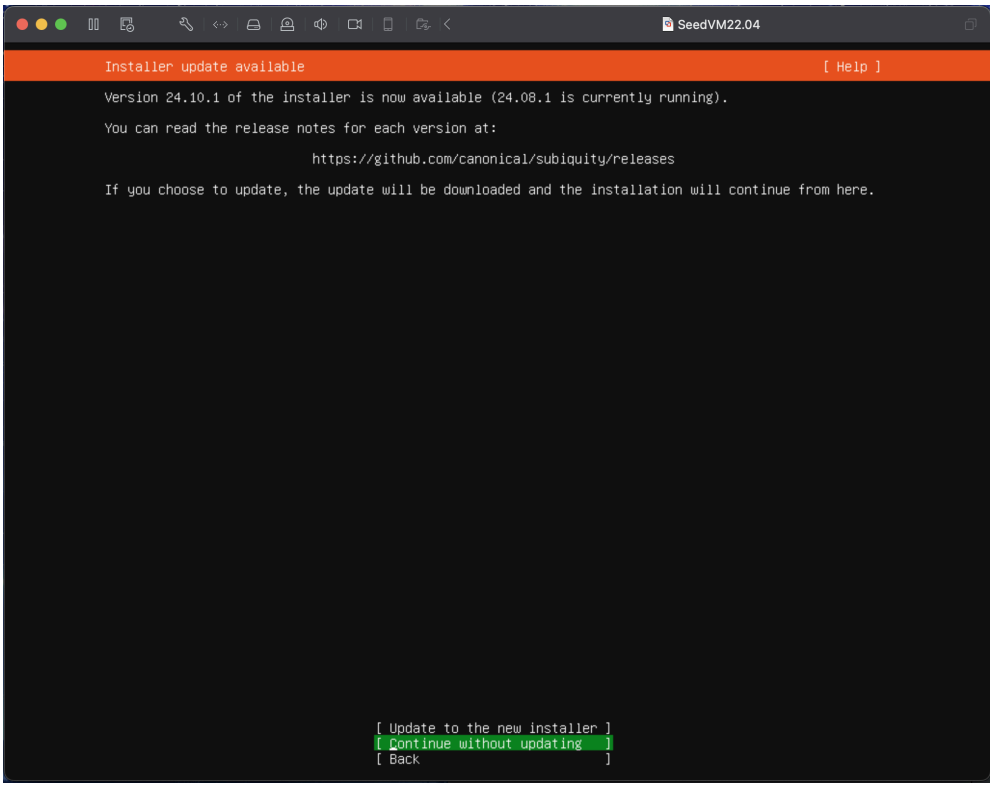


We will be greeted with the installation interface. Following the instructions below:

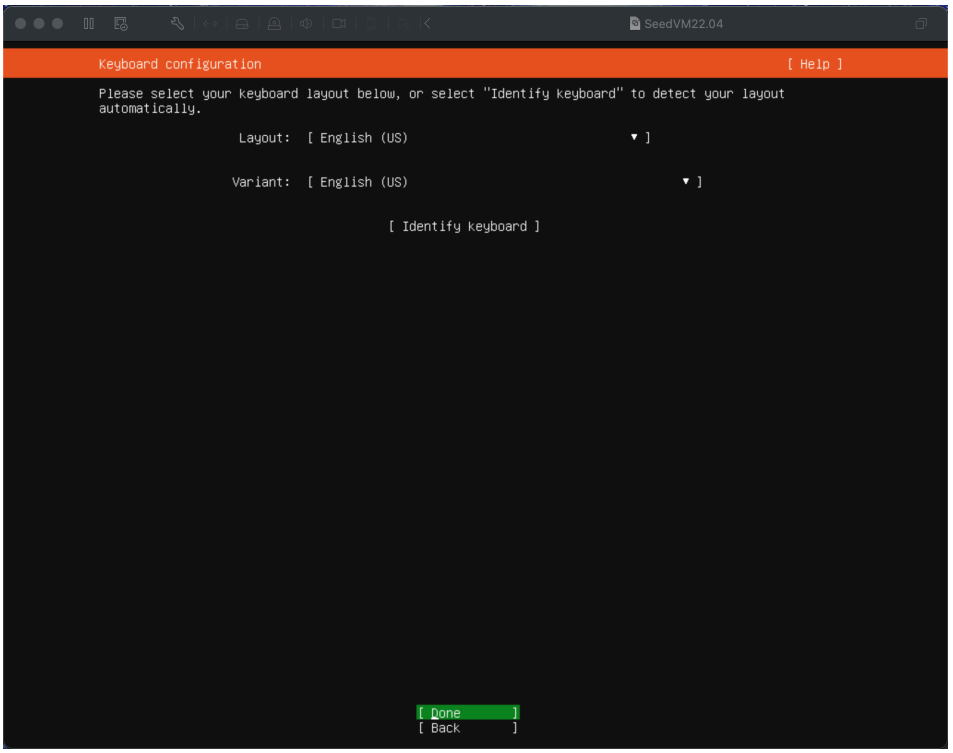
- Select the language as English.



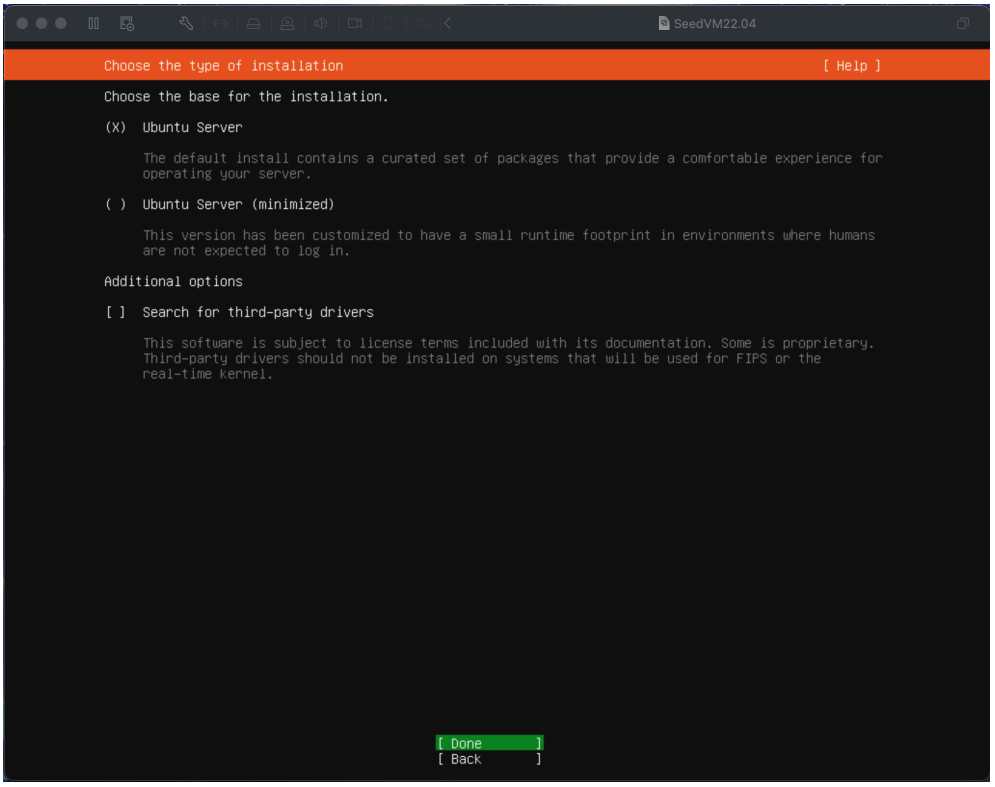
For available installer updates, select `Continue without updating`.



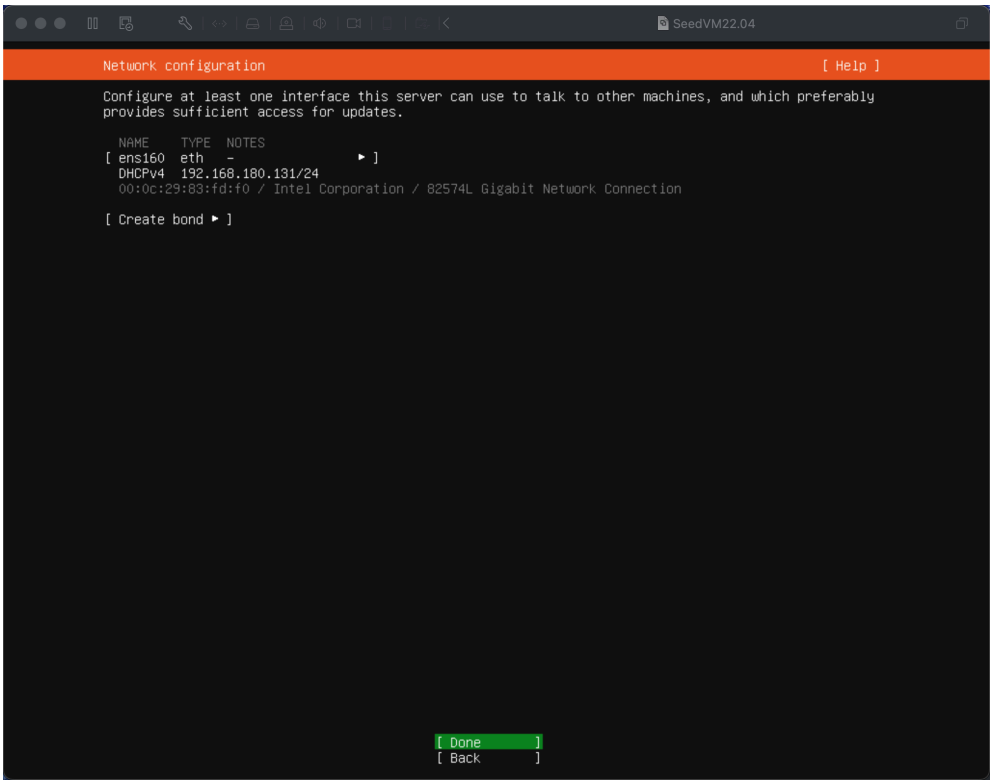
For Keyboard Configuration, select `Done` directly.



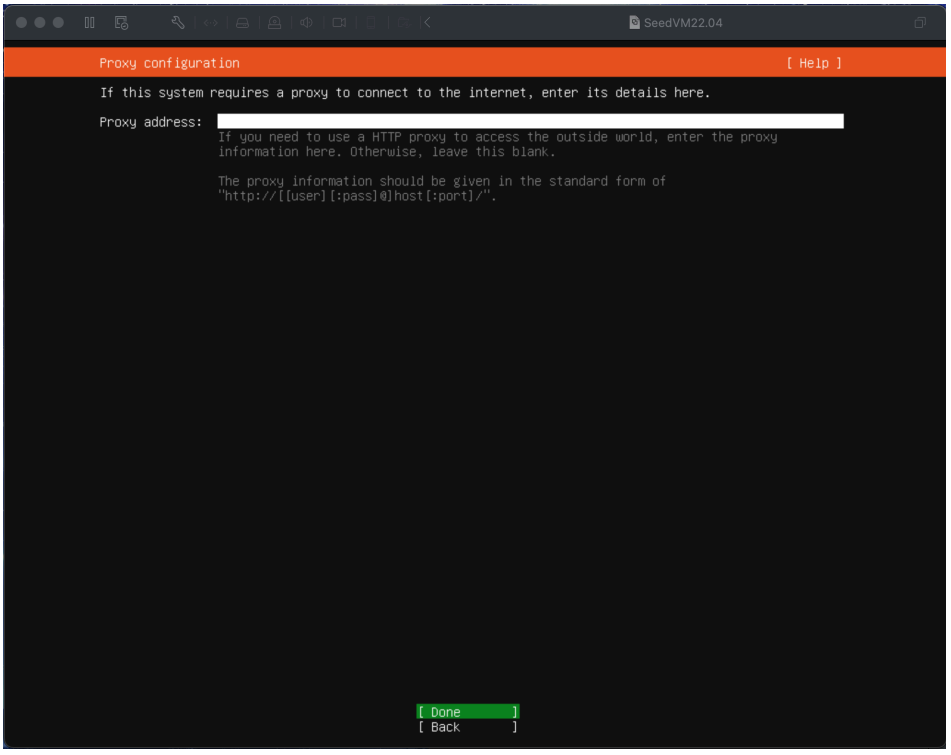
For Type of Installation, select `Ubuntu Server` and `Done`



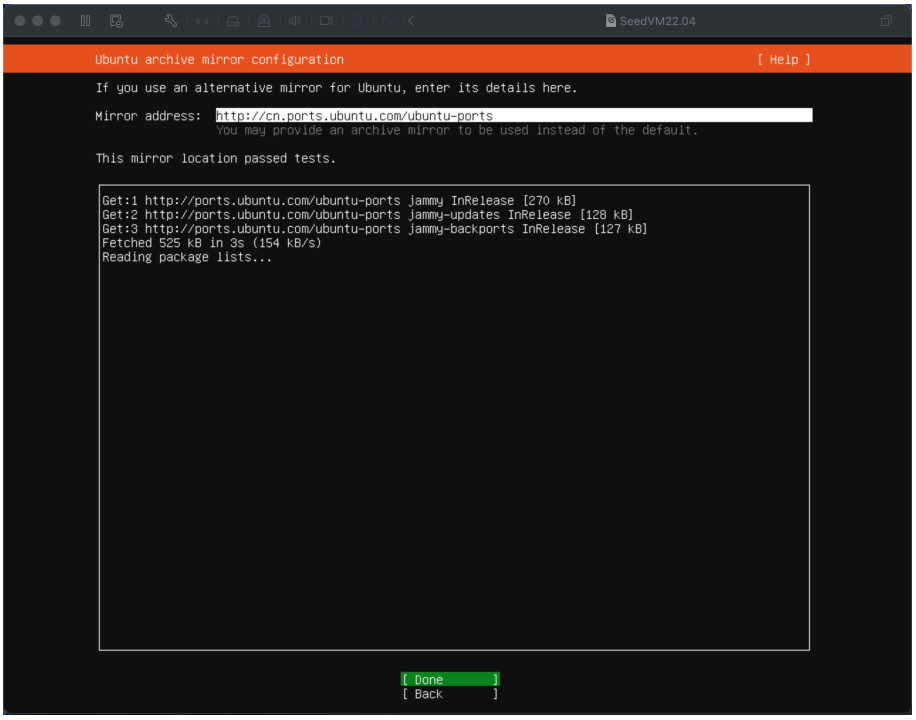
For Network Configuration, select `Done` directly.



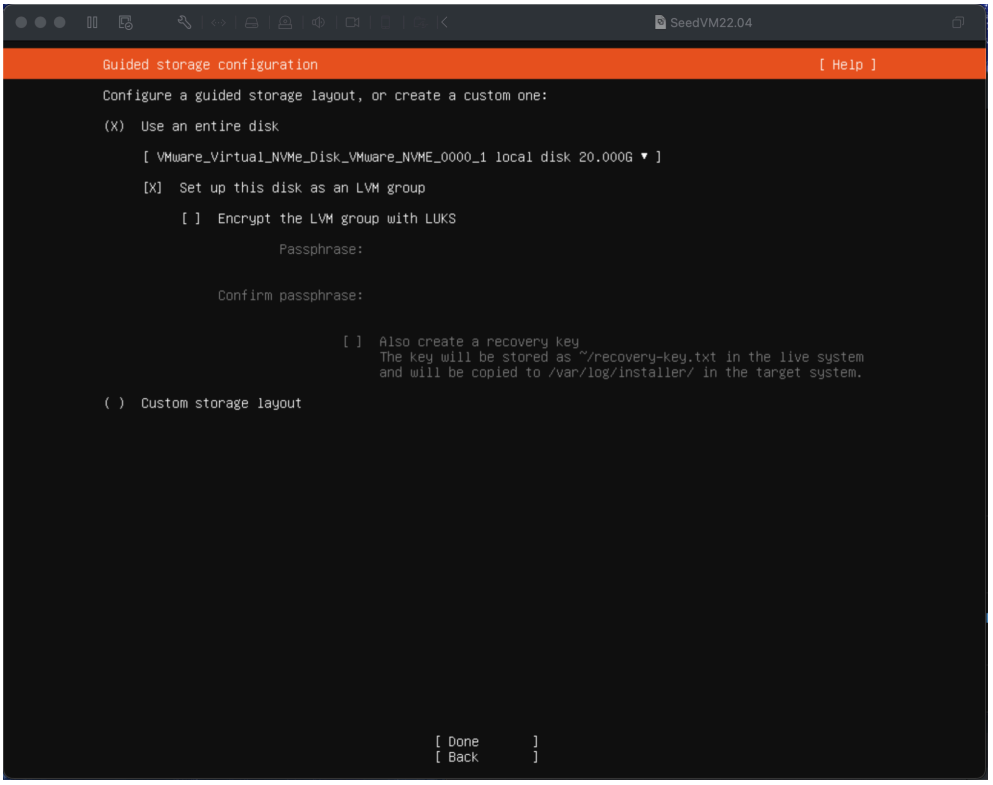
For Proxy Configuration, select `Done` directly.



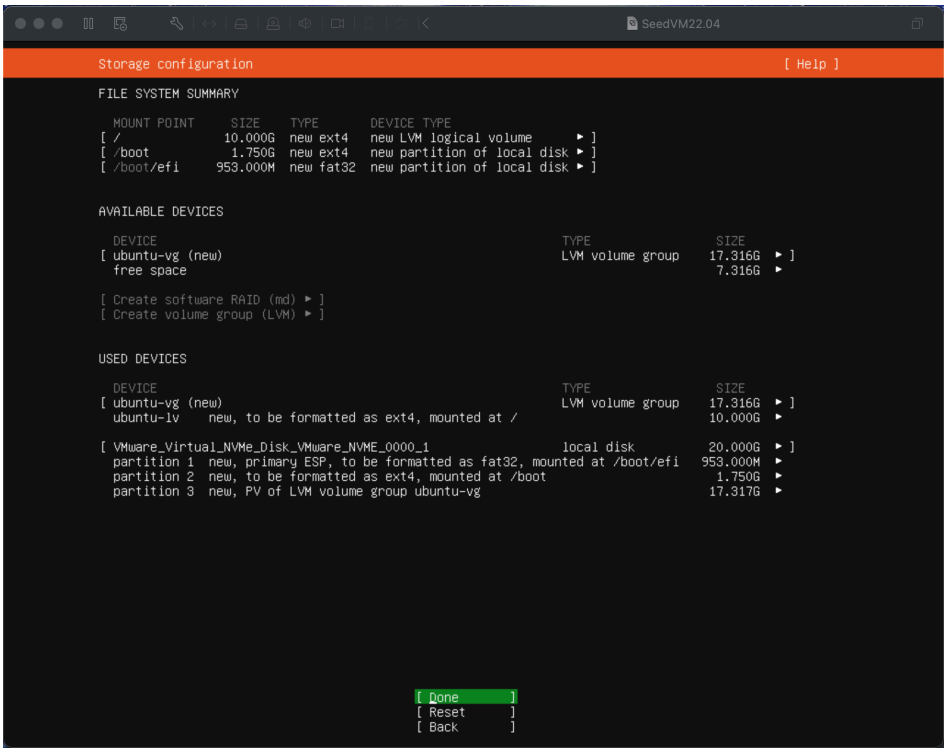
For Ubuntu Archive Mirror Configuration, wait until it completes the mirror location test and select `Done` directly.



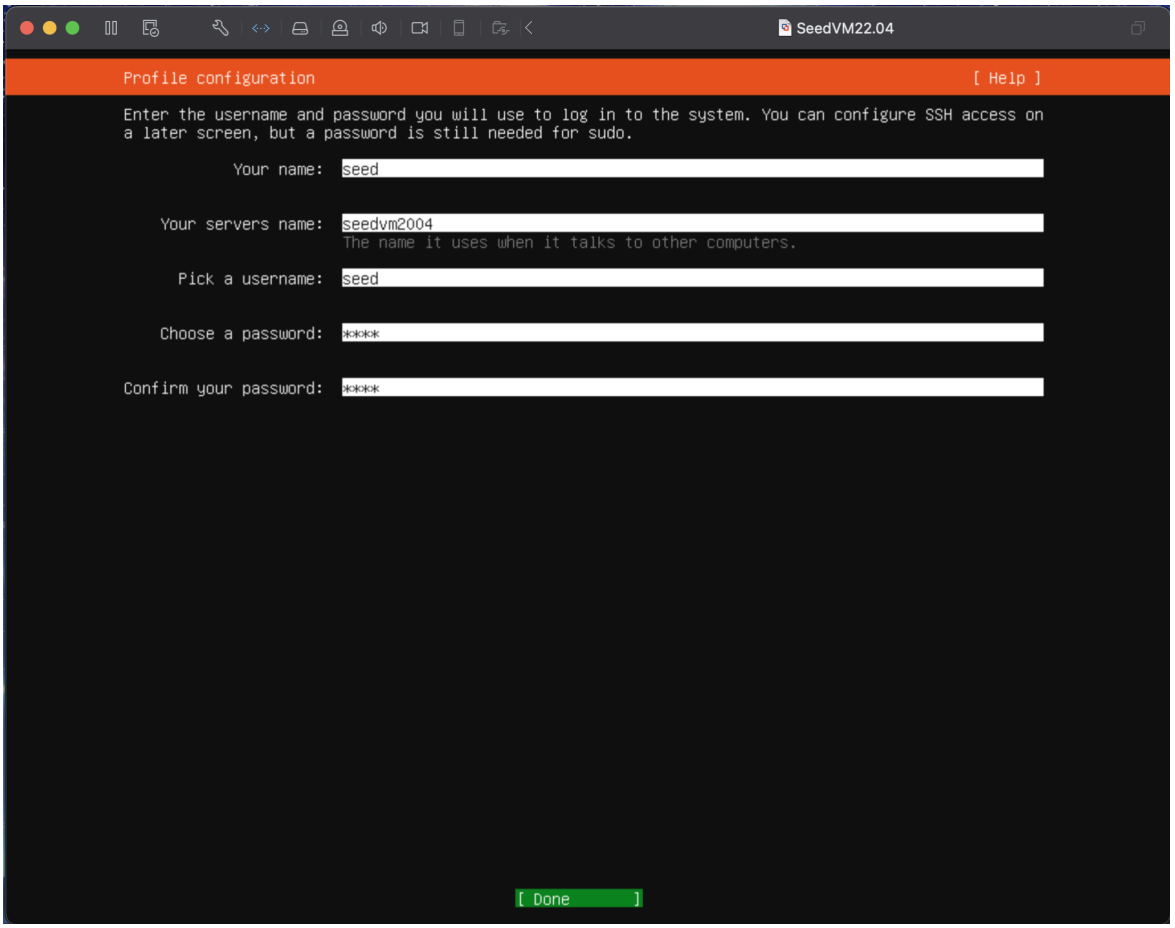
For Guided Storage Configuration, select `Done` directly.



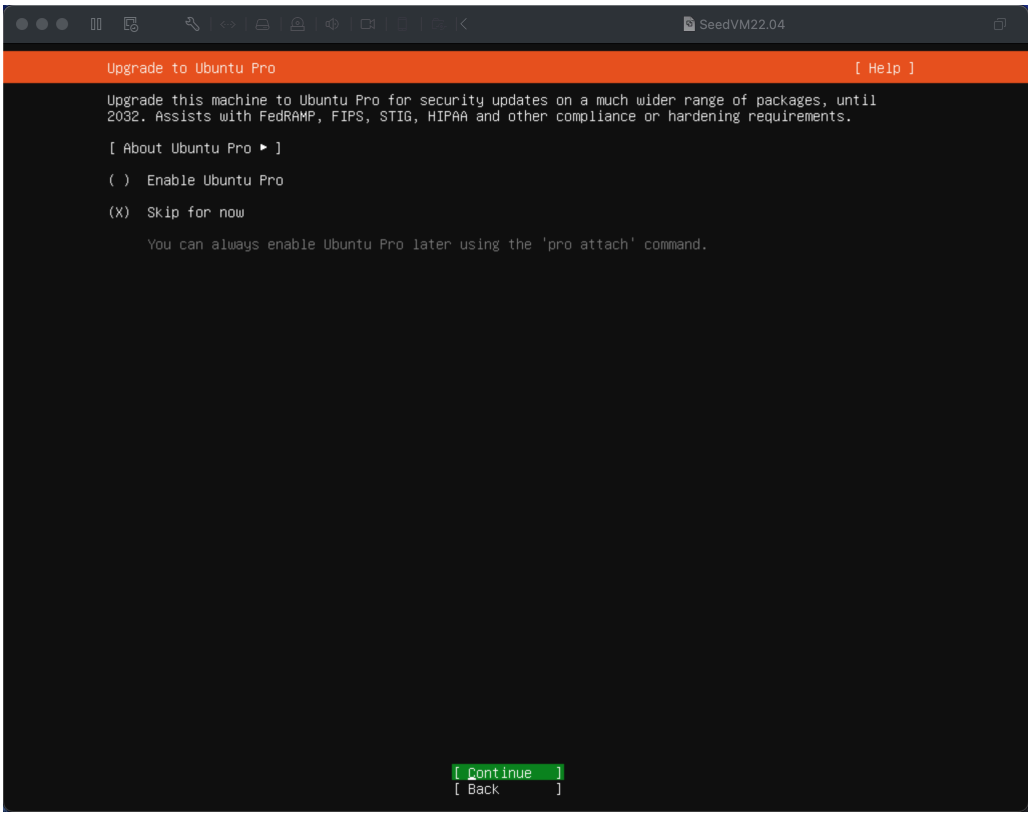
For Storage Configuration, select `Done` and `Continue` directly.



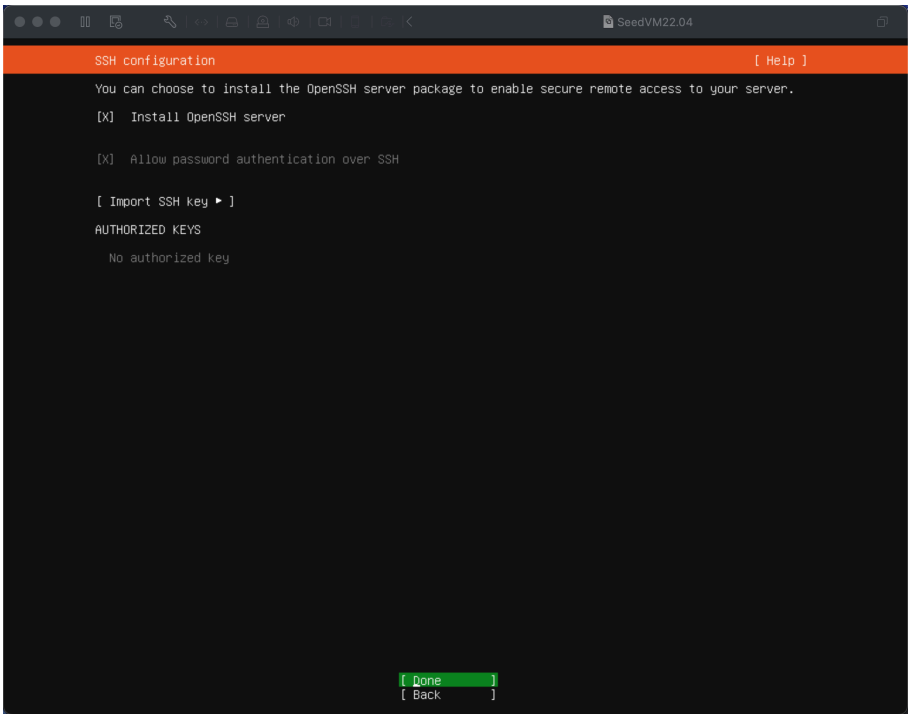
For Profile Configuration, create a user with name `seed` and select a password (you can use the standard password `dees` that we use for all SEED VMs). Then select `Done`.



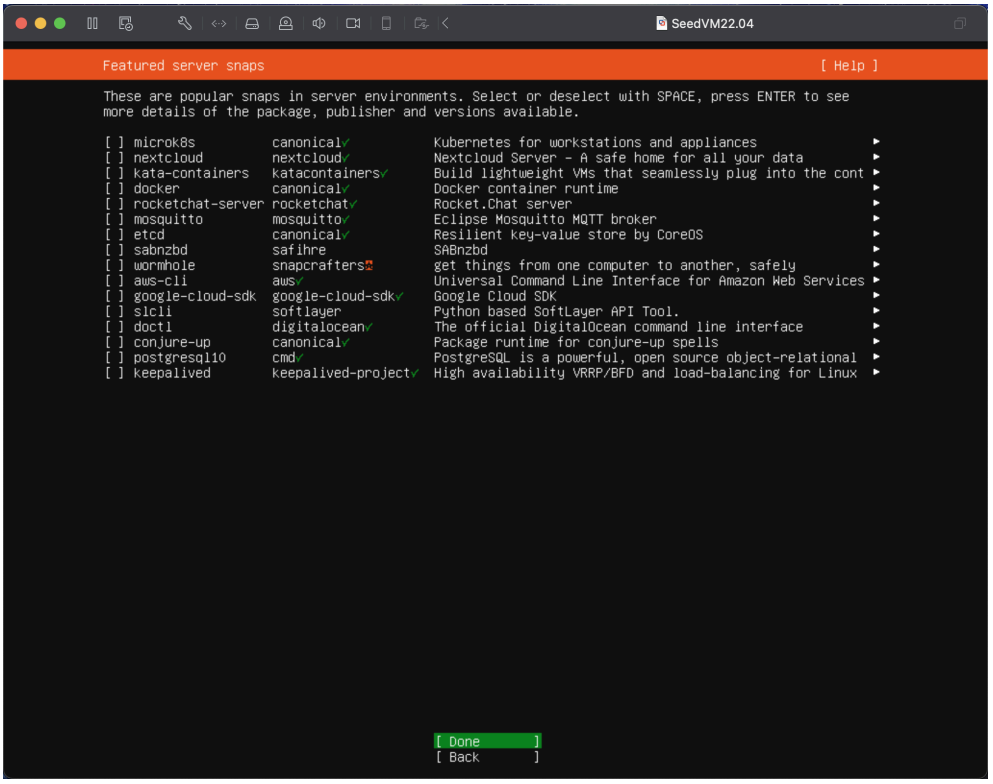
Select `Continue` directly.



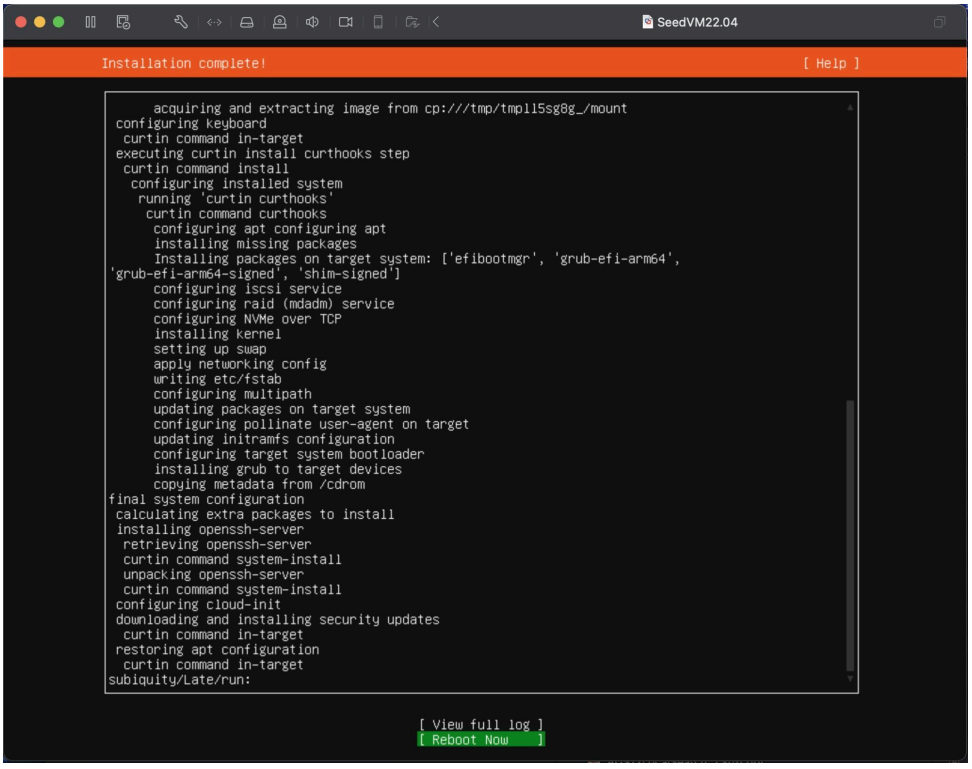
For SSH Configuration, we recommend you to enable it, so select `Install OpenSSH server` and `Done`.



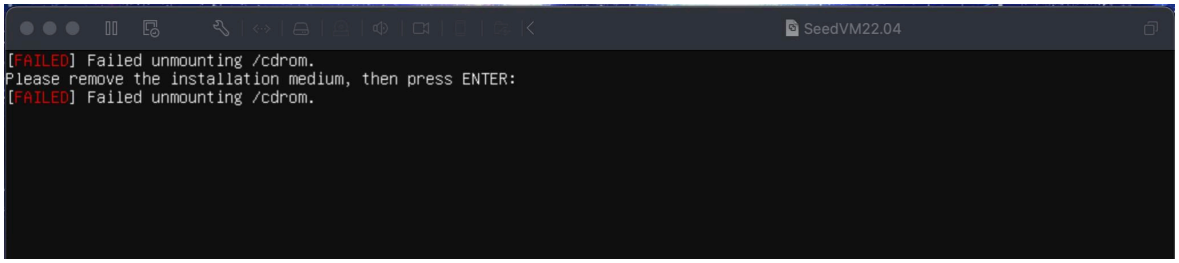
For Featured Server Snaps, select `Done` directly.

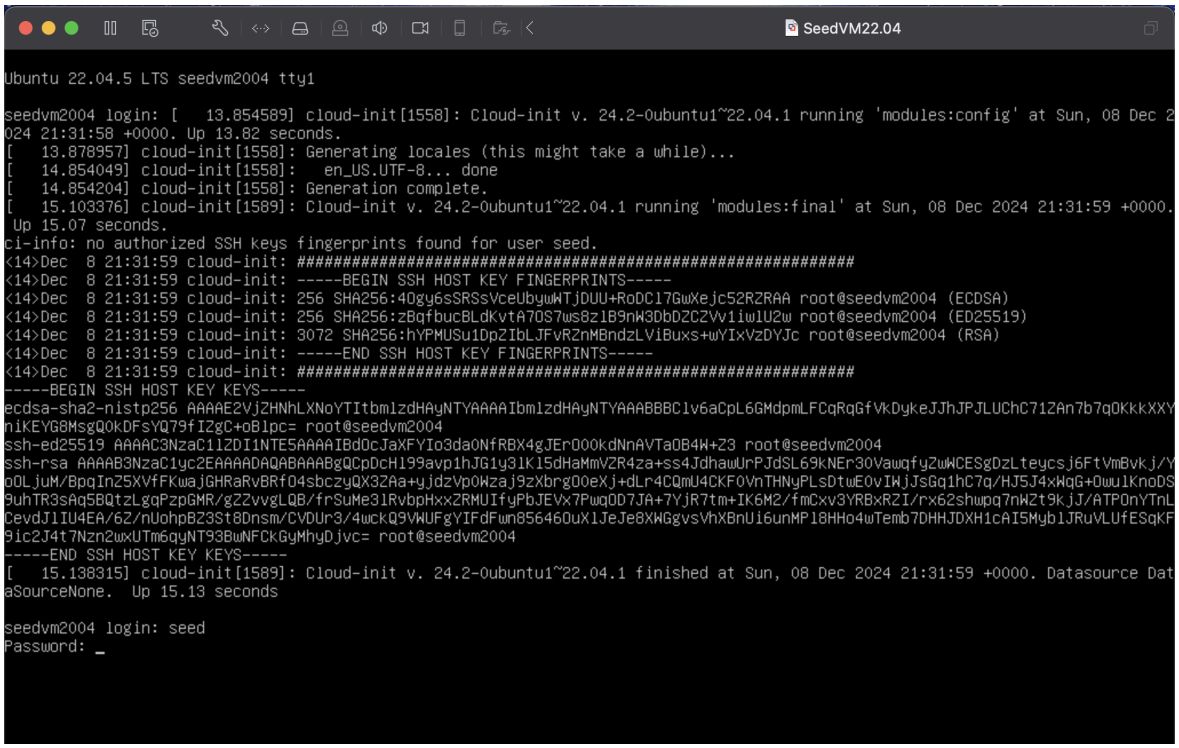


Wait until it finishes the installation and select `Reboot Now`



Maybe you will encounter a situation like `Failed unmounting /cdrom`, press enter directly.



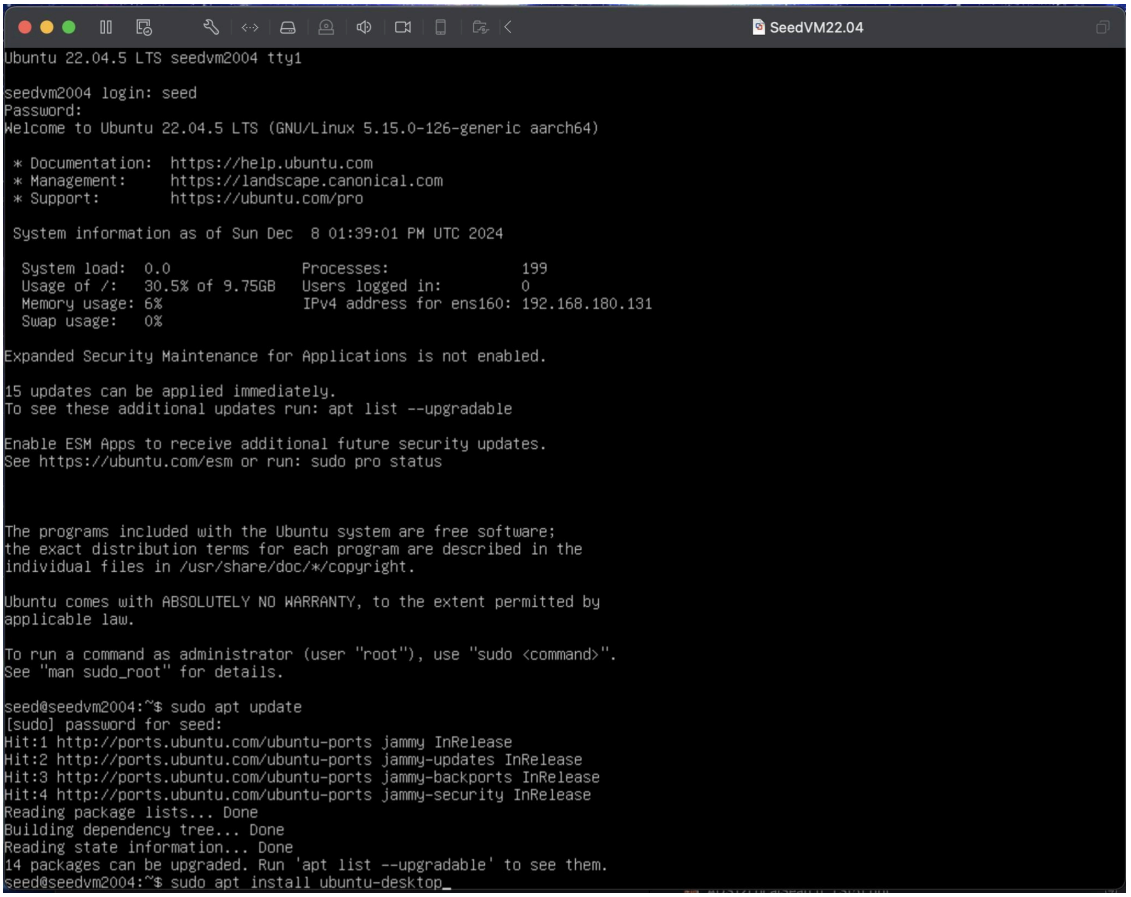


After the cloud-init operation, press enter and login to your account:

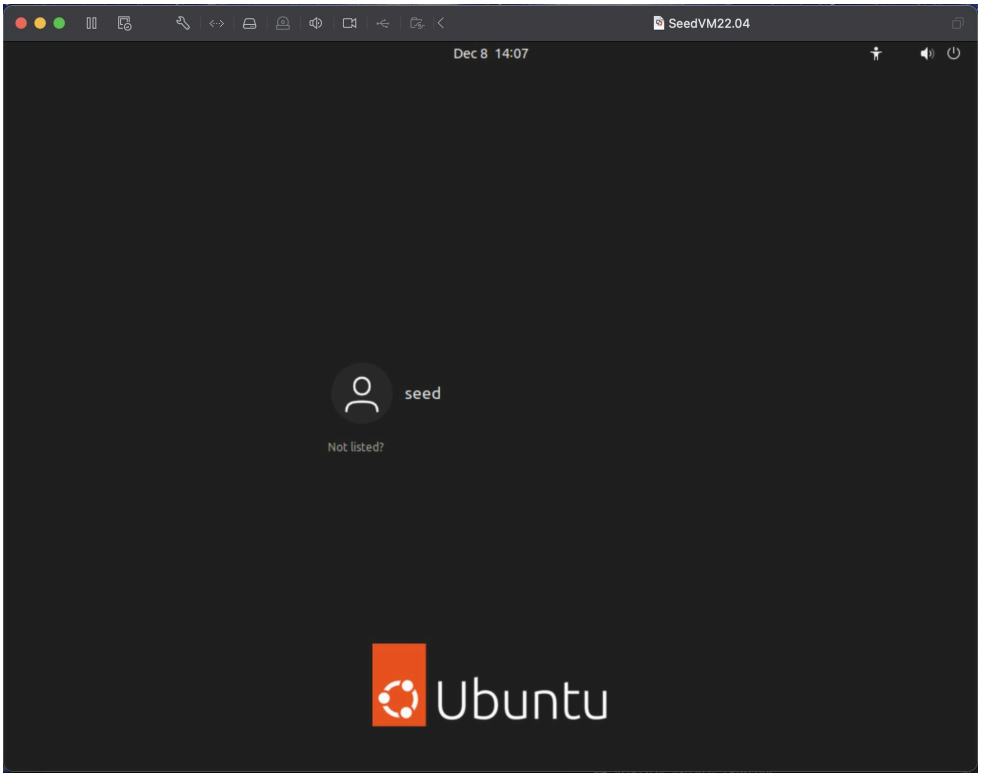
Step 2.4: Install Ubuntu Desktop

In most SEED labs, we need to have a GUI to check the result. We will install ubuntu-desktop in the VM to get it.

Use `sudo apt update` first and
`sudo apt install ubuntu-desktop`



Wait until it finishes the installation. This may take a long time. Then `sudo reboot` and we will get the GUI we want!



Step 3: Lab Setup

We are now ready to install all the software packages needed

Do not upgrade your Ubuntu version!

- Step 3.1: Setup Bidirectional Shared Clipboard
- Step 3.2: Install the Software

Step 3.1: Setup Bidirectional Shared Clipboard

In the default setting, the copy-and-paste does not work between your host Apple machine and the VM running inside VMware Fusion. This is quite inconvenient. You can install `vmware tools` to set up a bidirectional shared clipboard. Run the following commands.

```
sudo apt-get upgrade
sudo apt-get install open-vm-tools-desktop -y
sudo reboot
```

Step 3.2: Install the Software

Go to terminal, first download `curl` using

```
sudo apt-get install curl
```